THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS UNIT TEST SET B

Chapters 1–4: **Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations** | RBSE Board Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures; use graph paper where a graph is asked.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. The product of a non-zero rational number and an irrational number is: 1

(a) always rational **(b)** sometimes rational
(c) always irrational **(d)** always an integer

2. Two students find the HCF and LCM of 26 and 91. Student A gets HCF $= 13$, LCM $= 182$; Student B gets HCF $= 13$, LCM $= 2366$. Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

3. If $a = x^3y^2$ and $b = xy^3$, where x, y are primes, then HCF(a, b) is: 1

(a) x^2y^2 **(b)** xy^2 **(c)** x^3y^3 **(d)** xy

4. The sum of the zeroes of $2x^2 - 8x + 6$ is: 1

(a) −3 **(b)** 4 **(c)** −4 **(d)** 3

5. The maximum number of zeroes a cubic polynomial can have is: 1

(a) 3 **(b)** 4 **(c)** 2 **(d)** 1

6. If 2 is a zero of $x^2 - kx + 6$, then the value of k is: 1

(a) 3 **(b)** 6 **(c)** −5 **(d)** 5

7. The value of k for which $x + 2y = 3$ and $5x + ky + 7 = 0$ has no solution is: 1

(a) −10 **(b)** 2 **(c)** 5 **(d)** 10

8. The graph of the equation $x = 4$ is a line: 1

(a) parallel to the x-axis **(b)** parallel to the y-axis
(c) the y-axis itself **(d)** passing through the origin

9. The pair of equations $y = 0$ and $y = -7$ has: 1

(a) one solution **(b)** no solution
(c) infinitely many solutions **(d)** two solutions

10. The roots of $2x^2 - 5x + 3 = 0$ are: 1

(a) $2, 3$ **(b)** $-1, -\frac{3}{2}$ **(c)** $1, -\frac{3}{2}$ **(d)** $1, \frac{3}{2}$

11. If one root of $x^2 + px - 6 = 0$ is 2, then p is: 1

(a) 3 **(b)** −1 **(c)** 2 **(d)** 1

12. The discriminant of $3x^2 - 2x + \frac{1}{3} = 0$ is: 1

(a) 4 **(b)** 0 **(c)** −4 **(d)** 8

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

13. Find the HCF and LCM of 12, 15 and 21 by the prime factorisation method. **2**

14. If α and β are the zeroes of $x^2 - 5x + 6$, find the value of $\frac{1}{\alpha} + \frac{1}{\beta}$. **2**

15. Check whether the pair $5x - 4y + 8 = 0$ and $7x + 6y - 9 = 0$ is consistent. **2**

16. A student solved $x^2 - 2x - 15 = 0$ and wrote the roots as 5 and 3. Check whether $x = 3$ is a root and give the correct roots. **2**

17. Find k such that $kx^2 + 2x + 1 = 0$ has equal roots. For which values of k will it have no real roots? **2**

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

18. Prove that $6 + \sqrt{2}$ is irrational. **3**

19. Find a quadratic polynomial whose zeroes are $3 + \sqrt{2}$ and $3 - \sqrt{2}$. Verify your answer. **3**

20. Five years hence, the age of Jacob will be three times that of his son. Five years ago, Jacob's age was seven times that of his son. What are their present ages? **3**

21. Is it possible to design a rectangular park of perimeter 80 m and area 400 m²? If so, find its length and breadth. **3**

22. Draw the graph of $p(x) = x^2 - x - 6$ for $-3 \le x \le 4$ and find the zeroes of $p(x)$ from the graph. **3**

OR

Find a cubic polynomial with the sum, sum of the products of its zeroes taken two at a time, and the product of its zeroes as $2, -7, -14$ respectively. **3**

23. Find two consecutive odd positive integers, the sum of whose squares is 290. **3**

OR

The diagonal of a rectangular field is 60 m more than the shorter side. If the longer side is 30 m more than the shorter side, find the sides of the field. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

24. **Case study:** Traffic lights at three road crossings change after every 48 s, 72 s and 108 s respectively. They change simultaneously at 7:00 a.m. **5**

(i) Write the prime factorisation of 48, 72 and 108.

(ii) At what time will they next change simultaneously?

(iii) How many times will they change together between 7:00 a.m. and 8:00 a.m. (after 7:00 a.m.)?

OR

Prove that $\sqrt{7}$ is irrational. Hence prove that $2 + 3\sqrt{7}$ is irrational. **5**

25. A part of the monthly hostel charges is fixed and the rest depends on the number of days one takes food in the mess. Student A takes food for 20 days and pays ₹1000; Student B takes food for 26 days and pays ₹1180. Find the fixed charge and the cost of food per day. How much will Student C pay for 30 days? **5**

OR

A train covered a certain distance at a uniform speed. If the train had been 10 km/h faster, it would have taken 2 hours less; and if it had been 10 km/h slower, it would have taken 3 hours more. Find the distance covered. **5**

END OF QUESTION PAPER — Total: 50 Marks